

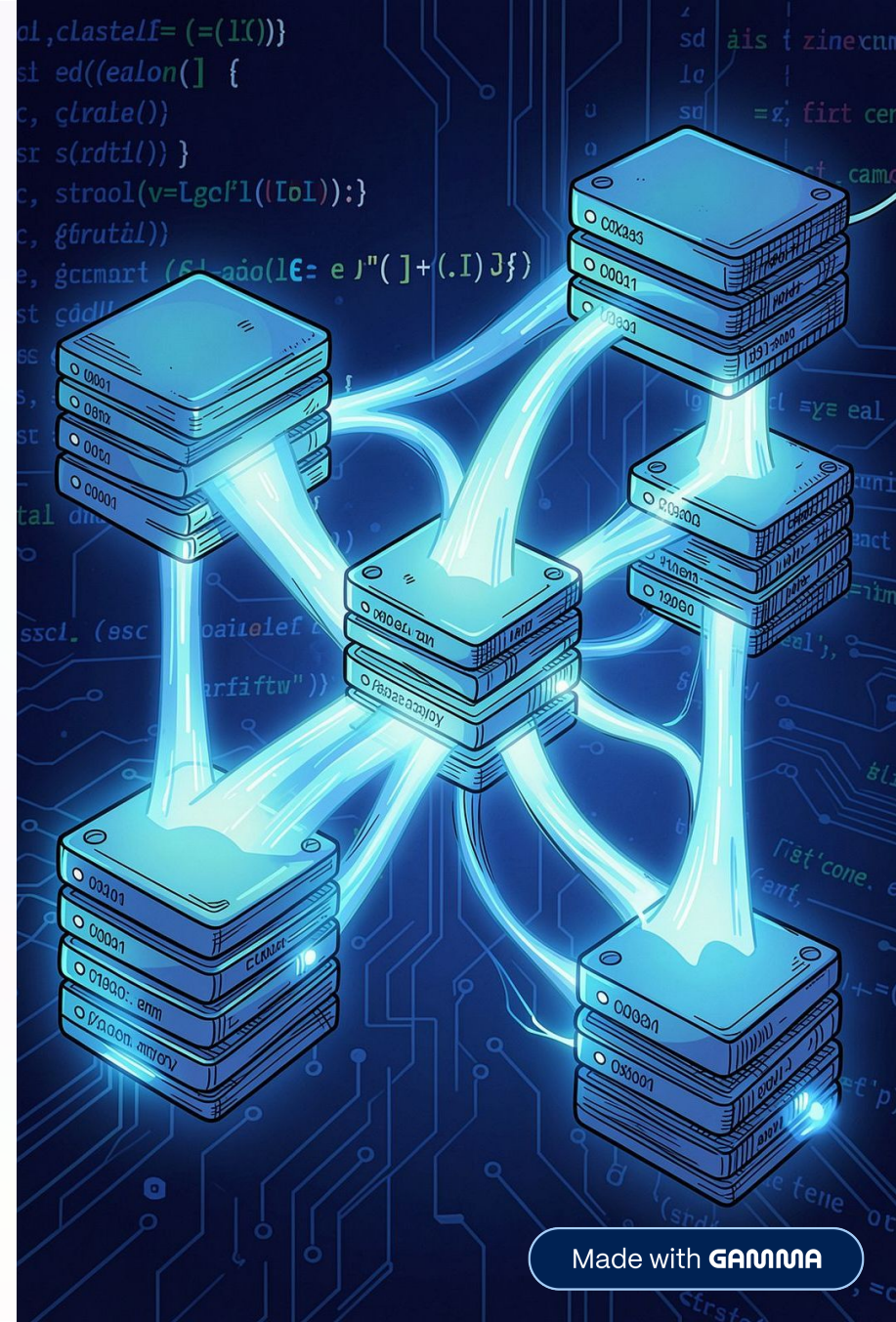
SQL Joins: Equi-Join vs Natural Join

Class XII CBSE Computer Science — Master the two most important join types for your board exams.

SQL

CLASS XII CBSE

DBMS



Why Joins Matter in SQL

Combine Tables

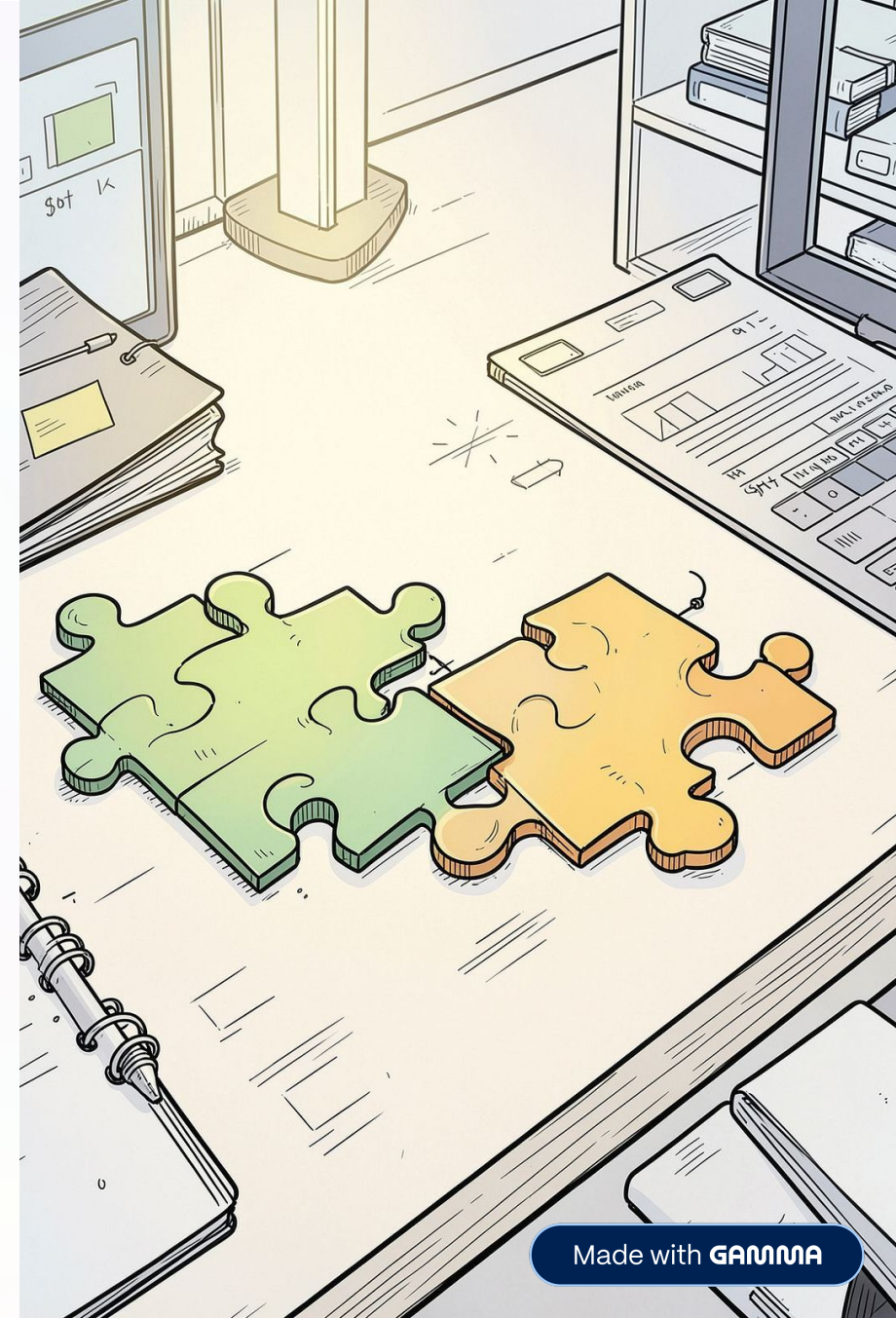
Joins merge rows from two or more tables into one meaningful result set.

Connect Related Data

Real-world data is split across tables — joins bring it back together.

CBSE Focus

Core topic: **"Retrieving data from multiple tables"** — frequently tested in board exams.



The Core Idea: Matching Rows

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A join uses a **join condition** — a criterion that decides which rows from each table get combined.

Join Condition

A predefined rule (usually equality) that must be satisfied for rows to appear in the output.

DBMS Terminology

In relational theory, joins combine **tuples** from two **relations** that meet the criterion.

Equi-Join: The Power of "="

An **Equi-Join** (also called Equality Join or Inner Join) combines two tables using an **equality condition** — matching rows where a column value in one table equals a column value in another.

1

Write the Condition

Use `WHERE table1.col = table2.col` or the `ON` clause.

2

Rows are Matched

Only rows satisfying the equality appear in the result.

3

Result Produced

Output includes all selected columns — including **duplicate common columns**.

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Equi-Join: Syntax & Output

Typical Syntax

```
SELECT columns  
FROM table1, table2  
WHERE table1.id = table2.id;
```

Or using explicit JOIN:

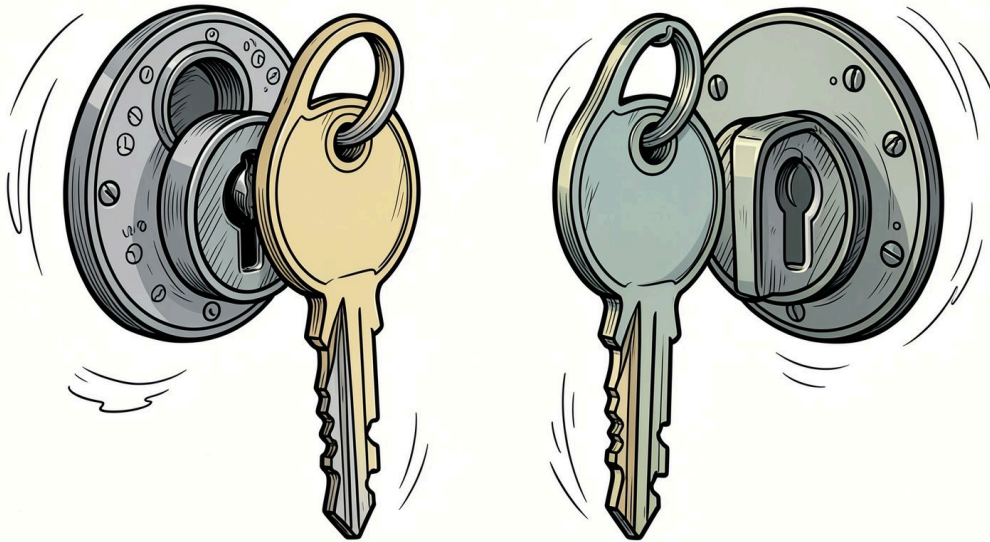
```
SELECT columns  
FROM table1  
JOIN table2  
ON table1.id = table2.id;
```

Key Exam Point

Equi-Join **does not automatically remove** duplicate common columns. If both tables have a column named `id`, both appear in the output.

-  You must explicitly choose which columns to SELECT to avoid redundancy.

Natural Join: Automatic Matching



A **Natural Join** is an inner join that automatically matches columns having the **same name and same datatype** in both tables — no explicit condition needed.

Implicit Join

No **ON** or **WHERE** clause — the database infers the condition.

Duplicate Elimination

Common columns appear **only once** in the result — a key difference from Equi-Join.

Natural Join: CBSE-Friendly Syntax

```
SELECT [column_names | *]  
FROM table1  
NATURAL JOIN table2;
```



No ON Clause

Never use **ON** or **USING** with **NATURAL JOIN** — it causes a syntax error.



One Copy Only

Each common column appears **once** in the output — automatic de-duplication.



Prerequisite

Matching columns must share the **same name AND same datatype** in both tables.

Similarities: Both Use Equality Logic



Equality-Based

Both rely on equality matching — Equi-Join uses `=` explicitly; Natural Join matches same-named columns implicitly.



Combine Records

Both are used to retrieve related records spread across two tables into a single result set.



CBSE Relevance

Both appear in typical Class XII questions: *"Write a query to display details from multiple related tables."*

Key Differences: Where Students Lose Marks



These differences are **frequently tested** in CBSE board exams. Remember: Natural Join is *implicit and automatic*, while Equi-Join gives you *explicit control*.

⚠ Most common mistake: Forgetting that Natural Join removes duplicate columns — always mention this in comparison questions!



Conclusion: Choose the Right Join

1 Use Equi-Join

When you need **explicit equality** on chosen columns — full control over the join condition.

2 Use Natural Join

When columns share the **same name and datatype** — automatic matching with duplicate elimination.

3 Master Both for CBSE

Be ready for **definition, comparison, and SQL writing** questions — both syntaxes are exam-critical.